USN | | | | | | | | | | [24MTAI11]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY 2025

MATHEMATICAL AND STATISTICAL FOUNDATIONS FOR AI

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	For what values of λ and μ, the system of equations $x+y+z=6$, $x+2y+3z=10$ and $x+2y+\lambda z=\mu$ has **i.** no solution, **ii.** unique solution and **iii.** infinite number of solutions.	**1**	**3**	**10**
	b.	Determine Eigenvalues and the corresponding Eigenvectors of the matrix $A=\begin{bmatrix}2&0&1\\0&2&0\\1&0&2\end{bmatrix}$. Also find the Matrix P such that $P^{-1}AP$ is a diagonal matrix if possible.	**1**	**3**	**10**
2.	**a.**	Fit a straight line for the following data that shows atmospheric pollutants y_i (relative to an EPA standard) at half hour intervals x_i:	**1**	**3**	**10**

x_i	1	1.5	2	2.5	3	3.5	4	4.5	5
y_i	-0.15	0.24	0.68	1.04	1.21	1.15	0.86	0.41	-0.08

	b.	Using LU decomposition method, solve the equations given below $x+y+z=1,\ 4x+3y-z=6,\ 3x+5y+3z=4$	**1**	**3**	**10**
3.	**a.**	Explain census and sampling methods in detail. How do they differ from each other?	**2**	**2**	**10**
	b.	If samples of size 2 are taken from the population 1, 2, 3, 4, 5, 6 without replacement, then find the mean and standard deviation of the population. Also find the mean and standard deviation of the sampling distribution of means.	**2**	**3**	**10**
		OR			
4.	**a.**	Describe the various probability sampling methods in detail and provide appropriate examples for each.	**2**	**2**	**10**
	b.	The weight of ball bearings proved in a certain factory are normally distributed with mean 0.5 N and standard deviation 0.002 N. Find the	**2**	**3**	**10**

Jain (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ram...

		percentage of ball bearings produced with weights **i.** less than 0.498 N, **ii.** more than 0.501 N, **iii.** between 0.4995 and 0.5005 N			
5.	**a.**	Define the following with suitable examples: **i.** Null hypothesis **ii.** Alternative hypothesis **iii.** Type-I and Type-II errors **iv.** Critical region and **v.** Level of Significance	**3**	**2**	**10**
	b.	From the two random samples data given below, Test whether the samples are selected from the same normal population. {Given $t_{5\%}(20) = 2.086$, $F_{5\%}(9,11) = 2.90$}	**3**	**3**	**10**

Sample	Size	Mean	Sum of squares of deviations from the mean
1	10	15	90
2	12	14	108

		OR			
6.	**a.**	Explain one-tailed test and two-tailed test in detail	**3**	**2**	**10**
	b.	Out of 400 men and 600 women, 200 men and 325 women were in favor of the flyover near their home. Test the hypothesis that proportions of men and women are same at 5% level. {Given $Z_{5\%} = 1.96$}	**3**	**3**	**10**
7.	**a.**	Outline the phases involved in solving an operations research problem.	**4**	**3**	**10**
	b.	Solve the following linear programming problem using graphical method. *Maximize* $Z = 40x + 80y$, *subject to* $2x + 3y \leq 48$, $x \leq 15$, $y \leq 10$ *and* $x \geq 0$, $y \geq 0$	**4**	**3**	**10**
		OR			
8	**a.**	Solve the following LPP using Simplex method $Max\ Z = 10x + 15y + 20z$ $Subject\ to$ $2x + 4y + 6z \leq 24$ $3x + 9y + 6z \leq 30$ $x, y, z \geq 0$	**4**	**3**	**12**
	b.	Solve the following LPP using graphical method: $Maximize\ Z = 5x + 3y$ $Subject\ to:$ $2x + y \leq 20$ $-4x - 5y \geq -10$ $x, y \geq 0$	**4**	**3**	**08**